

RESEARCH REPORT | REPORT 1

Reservoir Characterization and Formation Evaluation: Petrophysical Practice and Uncertainty

A technical review of log-derived saturation, core calibration, drive mechanisms, and reserves classification

Scope: conventional and tight clastic and carbonate reservoirs; industry-standard practice as codified in SPE-PRMS and API RP 40

Central conclusion

A reservoir model is only as defensible as the petrophysical inputs beneath it. Porosity, permeability, and saturation are not measured directly at reservoir scale; they are inferred through calibrated correlations carrying uncertainty that compounds into volumetric estimates. Formation evaluation should therefore report distributions and calibration provenance, not single deterministic values.

Executive summary

Formation evaluation converts indirect measurements into the parameters that govern hydrocarbon volume and flow. Wireline and logging-while-drilling tools respond to bulk physical properties: natural gamma radiation, electrical resistivity, bulk density, hydrogen index, and acoustic slowness. None of these is porosity or saturation. Each becomes a reservoir property only through a petrophysical model calibrated against core.

The dominant framework remains Archie's equation, published in 1942, which relates water saturation to formation resistivity through porosity and two empirical exponents. Archie's relation was derived from clean, water-wet sandstones with no conductive clay. Applied outside that envelope, most consequentially in shaly sands and heterogeneous carbonates, it systematically overestimates water saturation because clay-bound water provides a parallel conduction path the model does not account for. Shaly-sand models such as Waxman-Smiths and Dual-Water were developed specifically to correct this, and choosing between them is a material engineering decision rather than a formatting preference.

Uncertainty in these inputs propagates directly into volumetrics. Original oil in place is a product of several uncertain terms, so fractional errors accumulate approximately in quadrature. A 15% relative uncertainty in net pay combined with 10% in porosity and 20% in saturation yields roughly 27% uncertainty in OOIP before any consideration of recovery factor, which is itself the most uncertain term in the chain.

Evaluation principle

Report the calibration, not just the number. Every log-derived porosity and saturation curve should be traceable to the core dataset, the exponents fitted from it, the depth interval over which the fit holds, and the count of calibration points. An uncalibrated Archie curve is a hypothesis presented as a measurement.

Key findings

- Archie's exponents are fitted parameters, not constants. The cementation exponent m is near 2.0 in clean consolidated sandstone but ranges from roughly 1.3 in unconsolidated sand to well above 2.5 in vuggy or

moldic carbonate, where disconnected pore space raises resistivity without contributing to flow.

- The saturation exponent n is near 2.0 in strongly water-wet rock but rises substantially in oil-wet systems, where the conductive water phase loses continuity. Wettability is therefore an electrical parameter, not only a capillary one.
- Porosity and permeability are related only weakly and only within a defined rock type. Correlations fitted across pooled rock types are routinely misleading; permeability varies over orders of magnitude at fixed porosity depending on pore-throat geometry.
- Gas-measured permeability requires Klinkenberg correction for slip flow. The uncorrected value is systematically high, and the error grows as permeability falls, becoming severe in tight rock.
- Recovery factor, not volume in place, is the dominant uncertainty in most conventional developments. Drive mechanism ranges span roughly an order of magnitude in fractional recovery.
- SPE-PRMS reserves categories encode uncertainty explicitly. Reporting a single reserves number without its 1P/2P/3P context discards the information that makes the estimate usable.

1. Log measurements and what they actually respond to

Each logging measurement responds to a physical property that only partially overlaps with the reservoir property of interest. Understanding the gap between the two is the substance of formation evaluation.

The gamma ray tool records natural radiation from potassium, thorium, and uranium. It is conventionally read as a shale indicator, but the inference fails in two well-documented cases: potassium-feldspar-rich arkosic sandstones and glauconitic sands read as high gamma while being clean reservoir, and uranium salts precipitated from formation water produce hot streaks unrelated to clay content. The spectral gamma tool resolves this by separating the three contributions, which is why thorium-potassium crossplots are preferred for mineralogy over total gamma count.

Neutron and density porosity are complementary because their lithology errors run in opposite directions. The neutron tool responds to hydrogen index and therefore over-reads porosity in shale, where hydrogen is bound in clay structure, and under-reads badly in gas, where hydrogen density is low. The density tool converts bulk density to porosity through an assumed matrix and fluid density and is comparatively robust. Their combination produces the diagnostic gas crossover signature: neutron porosity suppressed below density porosity, one of the most reliable qualitative gas indicators available from open-hole logs.

Table 1. Principal open-hole logging measurements and their limitations

Measurement	Physical response	Primary use	Documented failure mode
Natural gamma ray	K, Th, U decay radiation	Shale volume; correlation between wells	Arkosic, glauconitic, and uranium-rich clean sands read falsely shaly
Spectral gamma	K, Th, U resolved separately	Clay typing; discriminating hot clean sand	Lower count-rate statistics; slower logging speed required
Bulk density (RHOB)	Compton scattering of gamma rays	Primary porosity via assumed matrix density	Barite mud invasion; rugose or washed-out hole; matrix assumption error
Neutron porosity	Hydrogen index of formation	Porosity; gas detection when paired with density	Over-reads in shale (bound water); under-reads badly in gas
Deep resistivity	Formation electrical conductivity	Saturation via Archie or shaly-sand model	Conductive clay; fresh formation water; thin-bed averaging
Sonic (DTC/DTS)	Compressional and shear slowness	Porosity; geomechanics; seismic tie	Cycle skipping; secondary porosity invisible to first arrival

Measurement	Physical response	Primary use	Documented failure mode
NMR	Relaxation of hydrogen spins	Lithology-independent porosity; bound vs. free fluid	Long acquisition; heavy-oil and viscosity-related response effects

Tool responses are averaged over a vertical resolution that is typically coarser than geological bedding. Thin-bed reservoirs require either high-resolution imaging or explicit deconvolution; standard-resolution logs will average pay and non-pay into an apparently uniform interval of intermediate quality.

2. Saturation modelling and the boundaries of Archie's equation

Archie's equation expresses water saturation raised to the saturation exponent as the product of a tortuosity factor and formation-water resistivity, divided by the product of porosity raised to the cementation exponent and true formation resistivity. Its physical premise is that the rock matrix is an insulator and that all electrical conduction occurs through connected brine in the pore network.

That premise carries the model's entire validity envelope. Where clay minerals contribute surface conductance, the measured resistivity is lower than the brine-only model predicts, and Archie interprets the extra conductivity as extra water. The result is pessimistic: pay is understated, and the error is largest exactly where it matters most, in laminated and dispersed-clay sands near economic cutoffs. Waxman-Smiths addresses this through cation exchange capacity per unit pore volume; the Dual-Water model instead partitions clay-bound water from free water as distinct conductive phases.

The tortuosity factor a is often set to 1.0 by convention, but the Humble formula and its variants use values below unity paired with a higher cementation exponent. Because a and m trade off against each other in the fit, quoting one without the other is meaningless. They are a matched pair from a single regression and must travel together.

Table 2. Archie parameter ranges by rock type

Rock type	Cementation exponent m	Saturation exponent n	Controlling factor
Unconsolidated sand	~1.3 to 1.6	~1.8 to 2.0	Grain contact geometry; minimal cement; well-connected pore network
Consolidated clean sandstone	~1.8 to 2.1	~1.8 to 2.2	Cementation degree; Archie's original calibration domain
Intergranular carbonate	~1.9 to 2.2	~1.8 to 2.2	Behaves broadly sandstone-like where porosity is intergranular
Vuggy / moldic carbonate	~2.2 to 3.0+	Highly variable	Isolated vugs raise resistivity without adding conductive path
Fractured carbonate	Can fall below 1.5	Variable	Fractures short-circuit conduction; m loses physical meaning
Oil-wet rock (any lithology)	Model-dependent	Can exceed 3 to 4	Discontinuous water film; conduction path loses connectivity

Ranges reflect values widely reported in the special core analysis literature and should be treated as priors pending field-specific core calibration, not as substitutes for it.

3. Porosity, permeability, and the limits of correlation

Porosity is a storage property and permeability is a transport property, and they are governed by different aspects of pore geometry. Porosity depends on total void fraction; permeability depends on the size and connectivity of pore throats, the constrictions between pore bodies. Two rocks at 20% porosity can differ by three orders of magnitude in permeability if one has coarse well-connected throats and the other has fine microporosity, which is why porosity-permeability transforms must be fitted within a single hydraulic flow unit

and are invalid when pooled across rock types.

Darcy's law defines permeability as the constant of proportionality between volumetric flux and pressure gradient, scaled by viscosity. It assumes laminar, single-phase flow of an inert incompressible fluid. Each assumption breaks somewhere in practice: high-rate gas flow near the wellbore becomes non-Darcy and requires the Forchheimer inertial term; multiphase flow requires relative permeability; and reactive fluids alter the pore structure they flow through.

Gas permeability measured in the laboratory requires Klinkenberg correction. Gas molecules slip at the pore wall rather than reaching zero velocity, producing higher apparent permeability than an equivalent liquid would give. The correction is obtained by measuring at several mean pressures and extrapolating to infinite pressure, where slip vanishes. The magnitude of the correction increases as permeability decreases, making it negligible in a darcy-scale sand and decisive in nanodarcy shale.

4. Drive mechanisms and recovery expectation

The natural energy that moves hydrocarbons toward the wellbore determines both the pressure history and the achievable primary recovery. Drive mechanism is diagnosable from the pattern of reservoir pressure, producing gas-oil ratio, and water production over time, and that diagnosis should precede any depletion plan.

Solution gas drive, in which gas evolving from the oil below bubble point provides the expansion energy, is both the most common and the least efficient conventional mechanism. Its diagnostic signature is a rapid pressure decline with a GOR that rises sharply once free gas becomes mobile, then falls as the reservoir depletes of gas. Because gas has far higher mobility than oil, it channels to the wellbore and leaves oil behind. Water drive, at the other end, can maintain pressure near-constant and displace oil with a far more favourable mobility contrast, but only if aquifer influx keeps pace with offtake; over-producing a water-drive reservoir converts it into a partial-depletion drive and forfeits the recovery advantage.

Table 3. Primary drive mechanisms, signatures, and typical recovery

Drive mechanism	Energy source	Diagnostic signature	Typical primary RF
Solution gas drive	Gas evolving below bubble point	Rapid pressure decline; GOR rises steeply then falls; little water	~5 to 30% OOIP
Gas cap expansion	Expansion of free gas cap	Slower decline; GOR rises steadily as cap advances	~20 to 40% OOIP
Water drive	Aquifer influx	Pressure near-maintained; water cut rises; GOR near-constant	~35 to 75% OOIP
Gravity drainage	Gravity segregation with relief	Low GOR in downdip wells; strongly dip-dependent	Up to ~80% OOIP
Rock and fluid expansion	Compressibility above bubble point	Very steep decline over a small pressure window	~1 to 10% OOIP
Combination drive	Two or more of the above	Mixed signature; requires material balance to partition	Intermediate; case-specific

Recovery ranges are broad industry benchmarks conditioned on rock and fluid properties, offtake discipline, and well placement. They characterise expectation, not entitlement, and are not substitutes for material balance or simulation.

5. Volumetrics and uncertainty propagation

Original oil in place is computed as bulk rock volume multiplied by net-to-gross, porosity, and hydrocarbon saturation, divided by the oil formation volume factor that converts reservoir barrels to stock-tank barrels. Each term carries uncertainty, and because the terms multiply, relative uncertainties combine approximately in

quadrature rather than additively. This is the quantitative argument against deterministic single-value reporting: the base case is not the mean, and the distribution is right-skewed because the product of positively distributed terms is approximately lognormal.

The formation volume factor deserves specific attention because it is frequently treated as a known constant. It is a PVT-derived property that varies with pressure and, at bubble point, reaches a maximum that can exceed 1.5 reservoir barrels per stock-tank barrel in volatile oils. Using a single B_o across a reservoir with significant pressure variation, or across compositionally distinct compartments, introduces a systematic bias in stock-tank volume that no amount of statistical treatment of the other terms will surface.

6. Reserves classification under SPE-PRMS

The Petroleum Resources Management System classifies recoverable volumes along two axes: commercial maturity and the degree of uncertainty. Reserves are the subset that is discovered, recoverable, commercial, and remaining, and each of those four conditions must hold simultaneously. Contingent resources are discovered and potentially recoverable but not yet commercial; prospective resources are undiscovered.

The probabilistic definitions are specific and frequently misquoted. Proved (1P) reserves correspond to at least a 90% probability that the quantity recovered will equal or exceed the estimate. Proved plus probable (2P) corresponds to at least 50%, and proved plus probable plus possible (3P) to at least 10%. The categories are cumulative, and their probabilities are not additive: incremental probable reserves do not carry a standalone P50. Reporting 2P as though it were a best estimate without stating the confidence basis strips out precisely the information the classification exists to convey.

7. Recommendations

- Fit Archie's exponents to field-specific special core analysis data and report a , m , and n as a matched set with the interval and sample count over which they were regressed.
- Test for clay conductivity before applying clean-sand Archie. Where cation exchange capacity is non-trivial, run Waxman-Smiths or Dual-Water and report the saturation difference against Archie as an explicit sensitivity.
- Fit porosity-permeability transforms within hydraulic flow units, and refuse pooled correlations across distinct rock types regardless of the apparent regression quality.
- Apply Klinkenberg correction to all gas-measured permeability and report both the corrected and uncorrected values with the mean pressures used in the extrapolation.
- Diagnose drive mechanism from pressure, GOR, and water-cut history before committing to a depletion plan; revisit the diagnosis as history accumulates rather than treating it as fixed at sanction.
- Report volumetrics probabilistically with stated input distributions, and state explicitly whether the quoted case is a mean, a mode, or a percentile of the output distribution.
- Carry SPE-PRMS category labels with every reserves number in every downstream document, including summaries prepared for non-technical audiences.

8. Limitations

This report summarises general petrophysical practice and does not substitute for field-specific core calibration, PVT analysis, or pressure-transient interpretation. Parameter ranges are drawn from widely reported literature values and vary substantially with depositional environment, diagenetic history, and wettability state. Recovery factor benchmarks are conditioned on development strategy and offtake discipline and should not be applied

as planning assumptions without local analogue support.

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